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CO1-AT-1-Short Answer Question

1. What is Data Science? Explain its primary objective.

Answer:

Data Science is the field of extracting useful knowledge and insights from data using statistics, programming, and machine learning.

Primary Objective:

- To analyze data and support better decision-making.

2. Why is Data Science important in today's digital world? Mention any four reasons.

Answer:

- Helps in better decision-making.
- Predicts future trends.
- Automates business processes.
- Improves customer experience.

3. List any four benefits of Data Science in business and society.

Answer:

- Better business decisions.
- Fraud detection.
- Improved healthcare.
- Personalized recommendations.

4. Mention four real-world applications of Data Science.

Answer:

- Healthcare diagnosis.
- Online recommendation systems.
- Banking fraud detection.
- Weather forecasting.

5. What are the different facets of data? Briefly explain each.

Answer:

- **Volume:** Amount of data.
- **Velocity:** Speed of data generation.
- **Variety:** Different types of data.
- **Veracity:** Accuracy and quality of data.
- **Value:** Useful information obtained from data.

6. Differentiate between structured, semi-structured, and unstructured data.

Type	Description	Example
Structured	Organized in rows and columns	Database, Excel
Semi-structured	Partially organized	XML, JSON
Unstructured	No fixed format	Images, Videos, Emails

7. What is Big Data? State its key characteristics (5 Vs).

Answer:

Big Data refers to extremely large datasets that cannot be processed using traditional methods.

5 Vs:

- Volume
- Velocity

- Variety
- Veracity
- Value

8. Explain the Big Data ecosystem. Name any four technologies used in it.

Answer:

The Big Data ecosystem consists of tools used to collect, store, process, and analyze large datasets.

Technologies:

- Hadoop
- Spark
- Hive
- HBase

9. What are the major phases of the Data Science process?

Answer:

1. Data Collection
2. Data Cleaning
3. Data Analysis
4. Model Building
5. Model Evaluation
6. Deployment

10. What is data retrieval? Mention two common sources.

Answer:

Data retrieval is the process of collecting data from different sources.

Sources:

- Databases
- APIs

11. Why is data collection considered an important step?

Answer:

- Provides quality data.
- Improves model accuracy.
- Supports better analysis.
- Helps in informed decision-making.

12. What is data cleansing? List any four techniques.

Answer:

Data cleansing is the process of removing errors and inconsistencies from data.

Techniques:

- Remove duplicates.
- Handle missing values.
- Correct errors.
- Remove outliers.

13. What is data integration? Why is it necessary?

Answer:

Data integration combines data from multiple sources into a single dataset.

Importance:

- Improves data quality.
- Eliminates redundancy.
- Enables better analysis.
- Provides a unified view.

14. Explain data transformation with suitable examples.

Answer:

Data transformation converts data into a suitable format for analysis.

Examples:

- Converting text to numbers.
- Normalization.
- Date format conversion.
- Scaling numerical values.

15. What is data analysis? Mention any four techniques.**Answer:**

Data analysis is the process of examining data to identify patterns and insights.

Techniques:

- Statistical analysis
- Data visualization
- Regression analysis
- Clustering

16. Differentiate between descriptive, predictive, and prescriptive analytics.**Analytics Purpose**

Descriptive What happened?

Predictive What will happen?

Prescriptive What should be
done?

17. What is meant by building a model? Mention the basic steps.**Answer:**

Model building is the process of training a machine learning algorithm using data.

Steps:

1. Select algorithm.
2. Train the model.
3. Test the model.

4. Improve performance.

18. Why is model evaluation important? Name any four evaluation metrics.

Answer:

Model evaluation measures the performance of a model.

Metrics:

- Accuracy
- Precision
- Recall
- F1-Score

19. Why is presenting findings important? Mention any four visualization tools or techniques.

Answer:

Presenting findings helps users understand insights and make decisions.

Tools/Techniques:

- Bar Chart
- Pie Chart
- Line Graph
- Tableau

20. How are Data Science models deployed into real-world applications? Give any four examples.

Answer:

Models are deployed through web applications, mobile apps, cloud services, or APIs for real-time predictions.

Applications:

- Recommendation systems
- Spam email detection
- Medical diagnosis

- Credit card fraud detection